

THEORYFORGE: OPERATOR-GUIDED AUTONOMOUS THEORY FORMATION IN MATHEMATICS

FROM THEOREM PROVING TO DEFINITIONS, INVARIANTS, REDUCTIONS, AND RESEARCH PROGRAMS

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ABSTRACT. Autonomous systems now emit plausible research-level mathematics, but emitting a statement is not discovery: novelty, truth, utility, and nontriviality are separate axes. We present THEORYFORGE, the *Discovery Plane* of the BOURBAKI ENGINE: an operator-guided architecture whose task is not to prove a supplied theorem but to *form theory* — to synthesize definitions, invariants, conjectures, reductions, counterexamples, barriers, and research programs whose value survives *independent* verification. THEORYFORGE generates typed *Mathematical Cells* through a library of thirteen discovery operators (e.g. SYNTHESIZE_DEFINITION, COUNTEREXAMPLE_FIRST, MINIMAL_MISSING_LEMMA, BARRIER_SEARCH), records their genealogy on an immutable event log, and — by construction, not by policy — holds *no promotion authority*: only an independent Congress may promote a claim. It reuses three existing substrates: MIRADOR (representation and content-addressed identity), PROOFCONTEXT (dependency-aware retrieval into hashed Evidence Bundles), and a CONGRESSBENCH-compatible reviewer interface. We define THEORYBENCH, a four-tier evaluation ladder (synthetic worlds, retrospective rediscovery, expert-authored unseen tasks, and a governed open frontier), a preregistered baseline/ablation protocol, and a five-stage novelty–verification gauntlet in which counterexample search precedes expensive proof promotion. The integrated case study is Bang’s affine plank conjecture, using a frozen, hash-pinned corpus of a submission-ready campaign.

 **Status.** This is a *design and protocol checkpoint*. The architecture, operator specifications, benchmark design, and evaluation protocol are written and grounded in the existing codebases; **no discovery experiment has been run, and no empirical result is reported**. Sections whose content depends on frozen-then-executed experiments are marked  **TO COMPLETE** and contain no fabricated numbers.

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Date: Checkpoint draft — July 12, 2026.

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1. INTRODUCTION

A model that prints an unfamiliar-looking statement has not made a discovery. Discovery requires that the statement be *true* (or that its falsity be a useful counterexample), *novel*, *nontrivial*, and *useful* to subsequent work — four axes that must be judged separately, and none of which is established by fluency. This paper concerns the machinery that produces candidate mathematics under exactly that discipline.

THEORYFORGE is the *Discovery Plane* of the BOURBAKI ENGINE [30], a governed architecture that separates two institutions: a Discovery Plane that *proposes* (and may be exploratory, redundant, or speculative) and a Trust Plane that alone may *promote* a claim into load-bearing knowledge. In the engine’s prior specification the Discovery Plane was described but not built (its maturity code is “conceptual”) [evidenced: Bourbaki audits/CLAIM_LEDGER.md]. THEORYFORGE is that missing implementation, together with a benchmark (THEORYBENCH) and an evaluation protocol. Five levels, two targets. We distinguish (i) *proof search* (derive a fixed claim); (ii) *conjecture generation* (propose statements); (iii) *program synthesis* (evaluator-optimized executable objects); (iv) *theory formation* (a *network* of definitions, claims, transformations, and explanatory structure); and (v) *research-program formation* (deciding which families of questions and representations deserve sustained effort). THEORYFORGE targets (iv) and (v), using (i)–(iii) as components. This is the line separating it from proof-only agents such as QED [18], RMA [19], Prover Agent [20], and LEAP [21], and from program-search discovery such as FunSearch [14] and AlphaEvolve [15], which optimize an executable against a fixed evaluator rather than forming a theory. Contributions (of this checkpoint).

- (1) A precise *theory-formation task* over a frozen, content-addressed corpus, with a typed output ontology of Mathematical Cells (§3).
- (2) A library of **thirteen discovery operators** with a common typed interface — input types, preconditions, transform, output state, validators, failure modes, and cost accounting (§4, App. A).
- (3) An **authority-separated architecture** that reuses MIRADOR, PROOFCONTEXT, and a CONGRESSBENCH-compatible reviewer, and in which THEORYFORGE structurally cannot promote (§5).
- (4) THEORYBENCH, a **four-tier evaluation ladder** and task ontology, and a preregistered baseline/ablation protocol (§§7–8).
- (5) A **five-stage novelty–verification gauntlet** and a multidimensional *Verified Knowledge Gain* record (§11).
- (6) A **frozen, hash-pinned case study** of Bang’s affine plank conjecture, separating rediscovery targets from genuinely open targets (§10, App. C).

Empirical evaluation (§9) is deferred until the protocol is frozen; this paper does not report it.

2. FROM THEOREM PROVING TO THEORY FORMATION

Automated mathematics has a long spine: the Logic Theory Machine [5], proof planning [6], the Robbins problem [7], and large formal verifications [8, 9, 10, 11]. *Theory formation* — inventing concepts, conjecturing, and finding counterexamples — was pioneered by AM [1] and systematized in HR [2], and remains an active target for the selection problem of “interestingness” [3]. What

is new is the confluence of (a) fluent large-model generation, (b) a governed, content-addressed knowledge substrate with deterministic invalidation [31], and (c) an explicit separation between proposing and promoting [30].

The methodological claim of THEORYFORGE is that *explicit discovery operators*, under a counterexample-first policy and a fixed information boundary, outperform unconstrained “think harder” prompting on theory-formation tasks — a claim this paper makes *testable* (§8) rather than asserts.

3. TASK DEFINITION AND OUTPUT ONTOLOGY

Task 1 (Theory formation). Given (a) a *governed corpus* $\mathcal{C}@v$ — a set of Mathematical Cells at a frozen content-addressed version v ; (b) a *frontier* $\mathcal{F} \subseteq \mathcal{C}@v$ — open questions and their dependency neighbourhood; (c) a *campaign constitution* K — exact conventions, admissible dependencies, per-claim-class certificate standards, novelty/attribution policy, and stopping conditions; and (d) a *budget* B split across six reserves (exploit, orthogonal explore, counterexample, formalize, barrier, reproduce), produce a set of typed Mathematical Cells, each created only in state [DRAFT], [OPEN], or [EVIDENCE], each carrying full genealogy and a fixed information boundary, maximizing *Verified Knowledge Gain* (§11) per unit cost — where “verified/novel” is decided by independent downstream machinery, never by THEORYFORGE.

Output ontology. Outputs are MIRADOR Mathematical Cells [31]: a canonical claim (IR statement plus its identity-bearing context Γ) bound to dependencies, transformations, artifacts, provenance, reviews, and an *epistemic state*. Two commitments are inherited and load-bearing [evidenced: MIRADOR `schema/mathematical-cell.schema.json`]: *object type* (one of twenty: DEFINITION, CONJECTURE, REDUCTION, COUNTEREXAMPLE, BARRIER, ...) is *orthogonal* to *epistemic state* (one of eleven: [DRAFT] $\rightarrow$ [OPEN] $\rightarrow$ [EVIDENCE] $\rightarrow$ [CANDIDATE] $\rightarrow$ [FORMALIZED] $\rightarrow$ [AUDITED] $\rightarrow$ [PROMOTED], plus [REFUTED], [NO_GO], [SUSPECT], [RETIRED]); and *claim identity* is the *SHA-256* of the statement and its context, never the prose surface, epistemic state, or artifacts. A cell that is merely reworded has the same identity; a renamed “discovery” therefore collides and is not novel (§11). THEORYFORGE emits cells no higher than [EVIDENCE].

4. DISCOVERY OPERATORS

Each operator implements a common typed interface (App. A; schema `schema/operator.schema.json`): `name`, `input_types`, `preconditions`, `transform`, `output_state`, `validators`, `failure_modes`, `cost`.

Every emitted candidate records its genealogy — operator, parents, Evidence-Bundle hash, model and prompt versions, and seed — and is serialized through MIRADOR so that identity is its `content_hash`. Output state is capped at [EVIDENCE]. Numeric witnesses are re-checked in *exact* arithmetic: a numeric grid is never a proof.

Table 1 summarizes the thirteen operators; four representative full specifications appear in Appendix A. Two operators (TRANSFER, CHANGE_REPRESENTATION) depend on a cross-domain analogy retrieval that PROOFCONTEXT does not yet expose (a v1 gap; §14, B2).

5. THEORYFORGE ARCHITECTURE

THEORYFORGE runs a closed but authority-separated loop: constitution $\rightarrow$ PROOFCONTEXT Evidence Bundle $\rightarrow$ frontier analysis $\rightarrow$ operator selection $\rightarrow$ candidate generation $\rightarrow$ exact experiments and counterexample search $\rightarrow$ formalization or certificate extraction $\rightarrow$ *independent Congress* $\rightarrow$ promoted result or structured failure $\rightarrow$ updated corpus and scheduler priors. The Congress step is the one-way membrane: every step before it may only *propose*; only the Congress promotes. Because THEORYFORGE writes through MIRADOR [31], whose transition table forbids an author

TABLE 1. The thirteen discovery operators.

Operator	Purpose (output object $\rightarrow$ max state)
SPECIALIZE	Restrict parameters/dimension/symmetry to expose tractable structure ($\rightarrow$ [OPEN]).
GENERALIZE	Identify which assumptions a proof actually uses; propose a broader object.
COUNTEREXAMPLE_FIRST	Attack a claim before allocating proof effort ($\rightarrow$ [EVIDENCE]).
NEGATE_AND_SEARCH	Compile negation to model/SAT/SMT/combinatorial search.
DUALIZE	Construct a dual formulation and an explicit correspondence obligation.
CHANGE_REPRESENTATION	Move among geometric, analytic, algebraic, combinatorial, probabilistic, logical, and computational forms.
TRANSFER	Import a pattern from another domain with a transport proof obligation.
MINIMAL_MISSING_LEMMA	Offer the weakest obligation that closes/advances a gapped blueprint.
EXTREMALIZE	Search for equality cases, extremizers, and rigid families.
STABILITY	Ask what near-extremal objects must resemble.
BARRIER_SEARCH	Prove a method family cannot cross a bound/obstruction ($\rightarrow$ [NO_GO] / [EVIDENCE]).
FORMALIZE_EARLY	Compile early to reveal hidden assumptions.
SYNTHESIZE_DEFINITION	Propose a concept that compresses repeated patterns; test if it improves statements, proofs, or retrieval.

from promoting its own cell and requires `actor=congress` with independence level ≥ 3 , the “no promotion authority” property is structural [evidenced: `MIRADOR transitions.check.transition`].

Contract 5.1 (Reused substrates). MIRADOR provides the cell IR, content-addressed identity, dependency graph, event log, and deterministic invalidation [31]. PROOFCONTEXT provides dependency-aware retrieval into a hashed Evidence Bundle [32] via `ProofContext.query(nl).bundle`. Independent review conforms to the CONGRESSBENCH **Reviewer** interface and `review.schema.json` [33]. THEORYFORGE depends on these interfaces, not their internals.

Modules. frontier mapper; operator policy; candidate generator; symbolic/computational laboratory (exact arithmetic); counterexample engine; proof planner; MIRADOR serializer; PROOFCONTEXT client; formal-backend adapter (see B1); CONGRESSBENCH-compatible reviewer interface; research-program scheduler (§6); and an immutable experiment/event logger.

6. RESEARCH-PROGRAM SCHEDULING

THEORYFORGE manages a *portfolio*, not a single chain of thought. Each program tracks: hypothesis, representation, target claims, known barriers, expected information gain, estimated cost, verification debt, novelty risk, dependencies, failed attempts, retreat conditions, and minimum publishable outcome. Explicit budgets are reserved for exploitation, orthogonal exploration, counterexample search, formalization, barriers, and reproduction. The scheduler score (a bandit / tree-search heuristic in the tradition of [23, 24]) allocates resources; **it is not evidence of mathematical truth**, and if a scalar is used its weights are preregistered (§11).

TABLE 2. Baseline/ablation matrix (arms $\times$ components).

Arm	Operators	PROOFCONTEXT	CE-first	Early-formalize	Scheduler
direct prompt	–	–	–	–	–
best-of- N	–	–	–	–	–
tree/search	–	–	–	–	–
proposer–critic	–	–	–	–	–
proof-only	–	–	–	–	–
program-search	–	–	–	–	–
TF –operators	–	•	•	•	•
TF –PROOFCONTEXT	•	–	•	•	•
TF –CE-first	•	•	–	•	•
TF –early-formalize	•	•	•	–	•
full TF	•	•	•	•	•

7. THEORYBENCH

Remark 7.1 (name). A name-availability check found no exact “THEORYBENCH” but a crowded neighbourhood (MathBench, UGMathBench, TPBench, GTBench, LemmaBench) [evidenced: web check 2026-07]. The name is provisional pending a less collision-prone choice.

THEORYBENCH spans four evaluation worlds. **Tier A — synthetic worlds:** small formal universes with hidden rules, concepts, and invariants; exact ground truth and full contamination control (the primary quantitative tier). **Tier B — retrospective:** time-slice a corpus before a known definition/lemma/bridge appeared and ask the system to reconstruct it; pretrained contamination is treated as a first-order threat and never called contamination-free. **Tier C — expert-authored unseen:** external mathematicians supply withheld tasks and criteria. **Tier D — open frontier:** bounded open subproblems under external expert governance, where success includes a new lemma, counterexample, reduction, barrier, or useful definition, not only full resolution.

Task families: definition synthesis, invariant discovery, conjecture ranking, counterexample generation, minimal-missing-lemma extraction, representation change, cross-domain transfer, extremal-family recovery, stability conjecturing, barrier discovery, research-program selection. Each task manifest fixes: allowed corpus @ v , allowed tools, hidden ground truth (where applicable), discovery budget, evaluation artifacts, novelty boundary, formal or expert checker, partial-credit rubric, and contamination risk.

 **TO COMPLETE.** *Concrete Tier-A synthetic universes (generators + checkers), the sealed Tier-C/D task set (stored hashed, released by a manifest event), and the per-family partial-credit rubrics are P5 deliverables. This section fixes the schema; the instantiated task bank is not yet built.*

8. EXPERIMENTAL PROTOCOL

Baselines (fixed budget, task-compatible only). Direct single-model prompting; best-of- N sampling; tree/search prompting [12]; proposer–critic agents; proof-only agents [18, 19]; and executable program-search [14].

Ablations of THEORYFORGE. –operator library; –PROOFCONTEXT; –counterexample-first policy; –early formalization; and full THEORYFORGE (Table 2). Comparisons are within a setting only; benchmark numbers are never compared across incompatible settings.

Freeze. Before any open-frontier run, five artifacts are frozen and hash-pinned as a MIRADOR event: (1) this protocol; (2) the Bang corpus manifest (App. C); (3) the THEORYBENCH task manifests

and sealed set; (4) the baseline/ablation matrix; (5) the Verified-Knowledge-Gain weights. Open-frontier runs are inadmissible until the freeze event exists.

9. RESULTS AND ABLATIONS

 **TO COMPLETE. No experiments have been run.** This section will report, per THEORYBENCH task family and per arm of Table 2, the multidimensional Verified Knowledge Gain of §11 at a fixed budget, with block-bootstrap confidence intervals and every number regenerable from raw artifacts under *results/*. Until the freeze (§8) and the resolution of the blocking decisions (§14), the table shells below are intentionally empty. Negative and null results will be reported, not suppressed.

TABLE 3. [Shell] Verified Knowledge Gain per arm, Tier A. *Empty pending frozen runs.*

Arm	Promoted	Refuted+CE	Barriers	Edges $\pm$	Compression	Cost
full TF	—	—	—	—	—	—
(baselines/ablations)	—	—	—	—	—	—

10. BANG AFFINE PLANK CASE STUDY

The problem. A *plank* of width w is the region between two parallel hyperplanes at distance w ; for a convex body K and a plank P with normal u , the relative width is $\text{rw}_K(P) = \text{width}(P)/\text{width}_K(u)$. *Bang’s affine plank conjecture* (1951) [25]: if K is covered by planks $P_1, \dots, P_n$, then $\sum_i \text{rw}_K(P_i) \geq 1$. Ball proved the centrally symmetric case [26]; the general case is open since 1951, open already in the plane, already for three planks [28], with best general lower bound $2/(1 + \sqrt{d})$ [28]. Gardner posed the relative-width witness-measure existence question [27].

The frozen campaign. The integrated case study freezes a submission-ready campaign on the triangle [29] (App. C). Its *proved* results — our **rediscovery targets** — include, all with exact-arithmetic backing [evidenced: `drafts/affine-plank-triangle.tex`]: a transport-defect framework ($D_K(u)$, the least uniform-marginal bound) giving $\sum \text{rw} \geq 1/D$; a *median theorem with full rigidity* (three median planks give $\sum \geq 1$, equality iff each is the central third); a *weighted-perimeter witness* (edge masses $1 - 2p_j$); tightness and rigidity of the cyclic family; a *three-direction characterization* (a uniform-marginal witness exists iff the three mid-level lines concur, settling Gardner’s question for three directions on the triangle [27]); a *normalizer obstruction* (Bang’s chord-lemma step is sharp, so that argument cannot reach any normalizer beyond the longest chord); and one *computer-assisted* result, $C_{\Delta}^{111}(\tau_0) = 1$ at the tilted triple $\tau_0 = (13/25, \frac{1}{2}, \frac{1}{2})$, discharged by an exact-rational emptiness certificate (812,651 leaves) re-verified by an independent checker [evidenced: `c3.sandwich.certificate.txt`, `bb.certificate.check.py`].

Rediscovery vs. novelty. The **novelty targets** are the genuinely open subproblems: the quantitative three-plank constant for non-concurrent, non-facet triples ($C_{\Delta}(u) = 1?$); the flagship *thin-plank lemma* (which would lift the finite branch-and-bound to the whole cyclic family); sufficiency of concurrence for $d+1$ directions on $\Delta^{d \geq 3}$; and the exact value of $\sup_u D_{\Delta}(u) \in [3/2, 2]$. Any allegedly novel Bang output must pass the five-stage gauntlet of §11.

Remark 10.1 (a naming discrepancy, reported not resolved). The project brief lists a “2F+1M” structure to reconstruct. No such object exists in the frozen corpus under any spelling [evidenced: `corpus grep`]; the closest are a “three facets + one plank” theorem and a “3 facet planks + 1 tilted” note. We do *not* invent a 2F+1M theorem; the intended target requires author clarification (§14, B4).

Remark 10.2 (contamination, stated plainly). The Bang manuscript is public (repository and journal submission) [evidenced]. Tier-B rediscovery on it therefore *cannot* be reported as contamination-free; rediscovered results are labeled rediscovery, never novelty. The corpus is frozen *with* poison flags on an unretracted-but-unsound note and on early refuted drafts, so that a run which “confirms” them scores as a failure (App. C).

 **TO COMPLETE.** *The THEORYFORGE discovery run over the frozen Bang corpus — rediscovery scoring, and any open-target attempt under external expert governance (Tier D) — is gated on the freeze and on blocking decisions B1–B7. No Bang discovery output is claimed in this checkpoint.*

11. NOVELTY, VERIFICATION AND PROMOTION

The five-stage gauntlet (counterexample-first). (1) PROOFCONTEXT *collision search* — `content_hash` plus `classify_identity` and `probable_similarity` over $\mathcal{C}@v$, and a literature audit for global novelty (must be built; the primitives exist in MIRADOR but PROOFCONTEXT exposes no collision client, B2). (2) *adversarial counterexample search before* any expensive proof effort. (3) *formal / certificate verification where feasible* — default tier: exact-rational/symbolic certificate + independent checker (the Bang standard, where the checker, not the search, is the authority); a proof assistant (Lean 4) is the named but not-yet-installed backend (B1, B6). (4) *fresh-context Congress* — a CONGRESSBENCH-conforming reviewer at condition $\geq C4$ run by ≥ 2 model families; the ACCEPT decision is the veto adjudicator’s, never THEORYFORGE’s. (5) *external expert review*, identified, not simulated.

Verified Knowledge Gain. A multidimensional record — promoted claims; refuted conjectures with reusable counterexamples; certified barriers; dependency edges added/removed; theory compression; external novelty status; cost and verification debt — *not* an opaque scalar. Quality axes (correctness, novelty, utility, compression, robustness, efficiency) are measured separately; “globally new” and “new-to-system” are distinct labels and never merged.

12. RELATED WORK

THEORYFORGE sits among automated theory formation (AM [1], HR [2], interestingness learning [3]), conjecture-from-data [16], program-search discovery [14, 15], auxiliary-construction geometry [13], proof-only research agents [18, 19, 20, 21, 17], and premise retrieval [22]. Its distinguishing commitments are the typed output ontology with orthogonal type/state, the propose/promote membrane, and a research-program memory across campaigns.

 **TO COMPLETE.** *The full comparison matrix (columns: fixed target, generates conjectures, synthesizes definitions, changes representation, finds counterexamples, finds barriers, formal verification, novelty audit, research-program memory) is a skeleton in `audits/RELATED_WORK_MATRIX.md`; its cells are unfilled pending a primary-source literature sweep. **No firstness is claimed** until that matrix and the collision-search protocol are complete.*

13. THREATS TO VALIDITY

We address, and design against: memorized discoveries; superficial renaming as novelty (caught by content-hash identity); false analogies; problem modification; reward hacking; evaluator overfitting; benchmark contamination; model monoculture (mitigated by ≥ 2 reviewer families); invalid transformation edges; computational evidence promoted as proof (blocked by the certificate/checker

standard); formalization of a weaker claim (tracked by `correspondence_status`); publication-priority errors; endless speculative branching and premature convergence (scheduler retreat conditions); excessive verification debt; and human-expert selection bias. The acute, project-specific threat is Bang contamination (§10).

14. LIMITATIONS AND RESEARCH AGENDA

This is a checkpoint: the architecture and protocol are specified, not executed. Eight decisions block the freeze and are stated for the record (recommendations given, not decided here):

- B1 Formal backend.:** Lean 4 is named but not installed [evidenced: CONGRESSBENCH B6]. Recommend adopting the exact-rational certificate + checker as the default formal tier; Lean optional/gated.
- B2 ProofContext gaps.:** No cross-domain analogy retrieval and no collision-search client [evidenced]; blocks TRANSFER/CHANGE_REPRESENTATION and gauntlet stage 1. Recommend building a thin collision wrapper over MIRADOR; descope cross-domain analogy to a v1 limitation.
- B3 Bang contamination.:** Public manuscript; keep Bang as an illustrative case with rediscovery labeled as such, and make Tier-A synthetic worlds the primary quantitative evidence.
- B4 2F+1M target.:** No such object exists; needs PI clarification.
- B5 Budget & model families.:** Reviewer independence needs ≥ 2 families at real dollar cost.
- B6 Frontier target & expert.:** Which open subproblem, and *who* is the identified external expert (manuscript authors? outreach thread?).
- B7 Congress composition.:** StubReviewer is not evidence; a real evaluative Congress needs LLM reviewers plus a human expert.
- B8 VKG scalar.:** Multidimensional for reporting; a preregistered scalar only for scheduler allocation.

15. CONCLUSION

THEORYFORGE operationalizes the distinction between emitting statements and forming theory, as the propose-only Discovery Plane of a governed engine. This checkpoint delivers the task definition, operator library, architecture, benchmark design, and verification protocol, plus a frozen case-study corpus — and, deliberately, *no discovery claims*. The next step is to resolve B1–B8, freeze the protocol and manifests, and run the preregistered evaluation, reporting Verified Knowledge Gain (including null results) that regenerates from raw artifacts.

APPENDIX A. DISCOVERY-OPERATOR SPECIFICATIONS (EXCERPT)

Four representative full specifications; the remaining nine follow the same seven-field schema in `operators/OPERATOR_API.md`.

COUNTEREXAMPLE_FIRST: *in:* CONJECTURE/STATEMENT. *pre:* falsifiable, with a searchable model class. *transform:* compile the negation to a search *before* any proof budget. *out:* COUNTEREXAMPLE ([EVIDENCE]) or a survived-search annotation. *val:* exact-arithmetic re-check of the witness. *fail:* search too large $\rightarrow$ bounded-search barrier note. *cost:* search wall-clock + calls.

SYNTHESIZE_DEFINITION: *in:* ≥ 2 EXAMPLE/EXPERIMENT/THEOREM sharing structure. *pre:* a retrievable repeated pattern. *transform:* propose a concept compressing it; test statement/proof/retrieval improvement. *out:* DEFINITION ([DRAFT]). *val:* IR type-check + non-collision. *fail:* pure renaming $\rightarrow$ flagged non-novel. *cost:* generation + compression measurement.

MINIMAL_MISSING_LEMMA: *in*: gapped PROOF blueprint. *pre*: localized gap. *transform*: infer the weakest closing obligation. *out*: LEMMA ([OPEN]) + **requires** edge. *val*: remaining steps type-check under the lemma. *fail*: lemma $\equiv$ theorem $\rightarrow$ flagged. *cost*: planner calls. (Bang: the thin-plank lemma.)

BARRIER_SEARCH: *in*: method family + target bound. *pre*: method formalizable as a bounded transformation. *transform*: prove it cannot cross an obstruction. *out*: BARRIER/[NO_GO] ([EVIDENCE]). *val*: exact/certified obstruction. *fail*: heuristic only $\rightarrow$ [EVIDENCE], not [NO_GO]. (Bang: the normalizer obstruction.)

APPENDIX B. OBJECT TYPES AND EPISTEMIC STATES

Object types (20, MIRADOR): CONCEPT, DEFINITION, CONSTRUCTION, STATEMENT, QUESTION, CONJECTURE, LEMMA, THEOREM, EXAMPLE, COUNTEREXAMPLE, REDUCTION, EQUIVALENCE, BARRIER, PROOF, REFUTATION, EXPERIMENT, CERTIFICATE, CHECKER, HEURISTIC, TRANSLATION. Epistemic states (11): [DRAFT], [OPEN], [EVIDENCE], [CANDIDATE], [REFUTED], [NO_GO], [FORMALIZED], [AUDITED], [PROMOTED], [SUSPECT], [RETIRED]. THEORYFORGE emits only [DRAFT]/[OPEN]/[EVIDENCE].

APPENDIX C. BANG CORPUS FREEZE (V0 DRAFT)

A read-only manifest of the frozen campaign was generated at `case_studies/bang/CORPUS_FREEZE.manifest.json`: 269 files, $\approx$ 32.9 MB, from the repository `maximilianolucius/bang-plank-affine-tri` at commit `68d176cd`, with rolling manifest hash `sha256:2ef41dd2...` [evidenced: generated 2026-07-12]. *Excluded*: the abandoned Euler–Jacobi drafts (`drafts/obsolete/`), an unrelated Hamilton–Jacobi note, and copyright-excluded reference PDFs. *Poison-flagged* (frozen *with* a warning, contamination control): an unretracted-but-unsound note and early refuted drafts. The file boundary and poison policy await PI sign-off before this becomes `bang-freeze-v0` (B-freeze).

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